

Monitoring and Modeling of Effluent Plume Dynamics and the Atmospheric CO₂ Drawdown Potential of Captura's Direct Ocean Capture System

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Large-scale marine Carbon Dioxide Removal (mCDR) is a promising climate mitigation strategy. Captura Corp employs a Direct Ocean Capture (DOC) process. To evaluate the efficacy and environmental impact of Captura's DOC system at scale, we utilize high-resolution observational data with advanced numerical modeling to project CDR drawdown at near-field scale in the Port of Los Angeles, CA and regional scale using historical data in the Southern California Bight.

Background and Overview

Combining near-field observations with regional models, our approach seeks to address the complexities of quantifying CO₂ drawdown from ocean-based CDR at scale. Validation and refinement of these models will enhance understanding of the environmental impacts and scalability of DOC.

Near-Field Modeling

In collaboration with Environmental Dynamics Lab (END Lab) at MIT, we monitor the near field dispersal of decarbonized seawater effluent in the Port of Los Angeles. Using in-situ sensors and field measurements, we capture the effluent plume's immediate dispersal patterns and mixing dynamics. This data can inform a high resolution, near field model of the plume dynamics.

Air-Sea CO₂ Flux Attributable to DOC

At the regional scale, the Southern California Coastal Water Research Project (SCCWRP) employs the Regional Ocean Model System (ROMS) coupled with Biogeochemical Elemental Cycling (BEC) to simulate the carbon removal and air-sea CO₂ flux of DOC in the Southern California Bight. This sub mesoscale-resolving model tracks the evolution of CO₂-depleted seawater and its re-equilibration with atmospheric CO₂ to determine the notable additionality of Captura's DOC interventions compared to a baseline scenario.

Captura Direct Ocean Capture Process

Captura's Direct Ocean Capture process

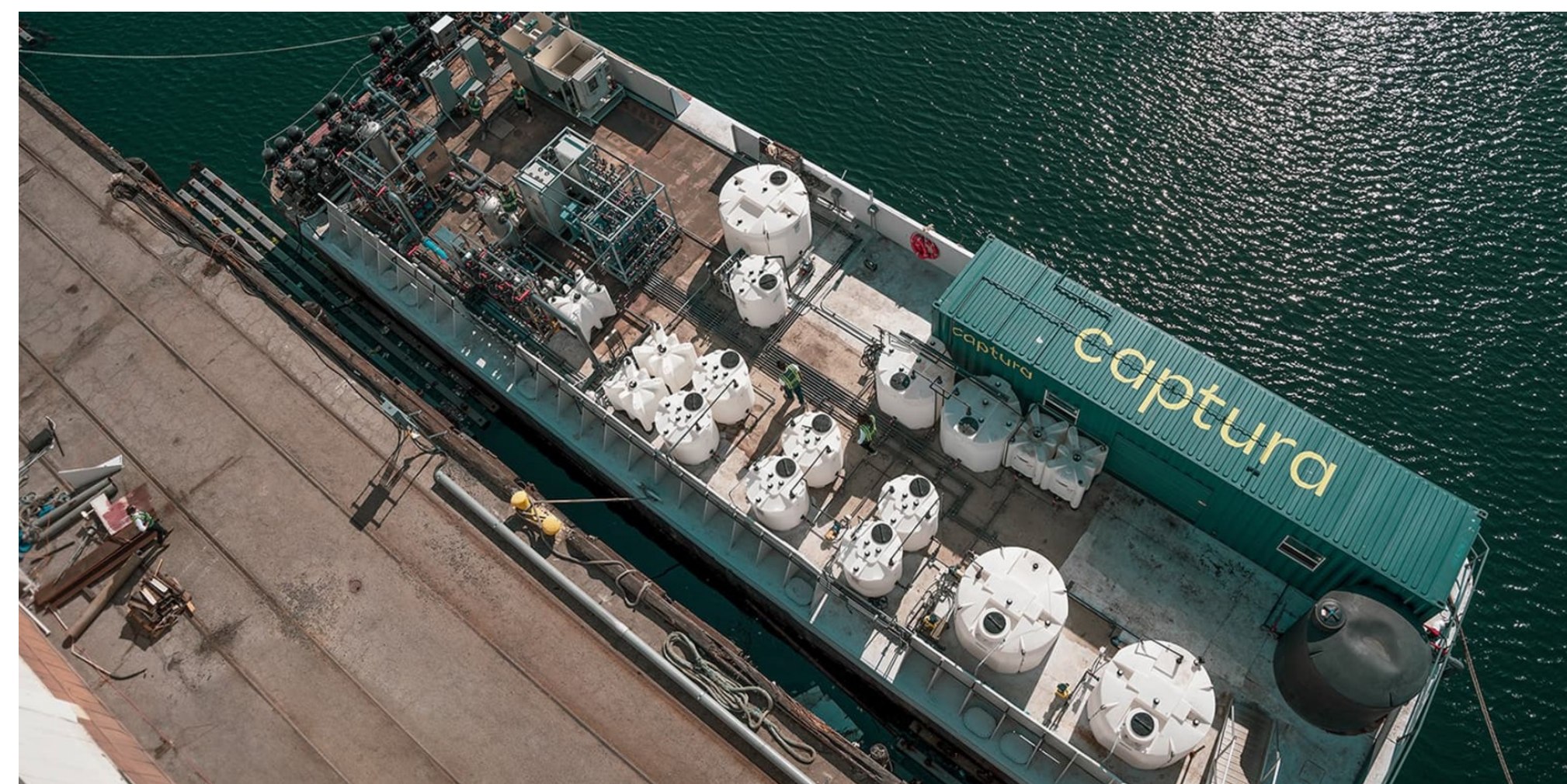
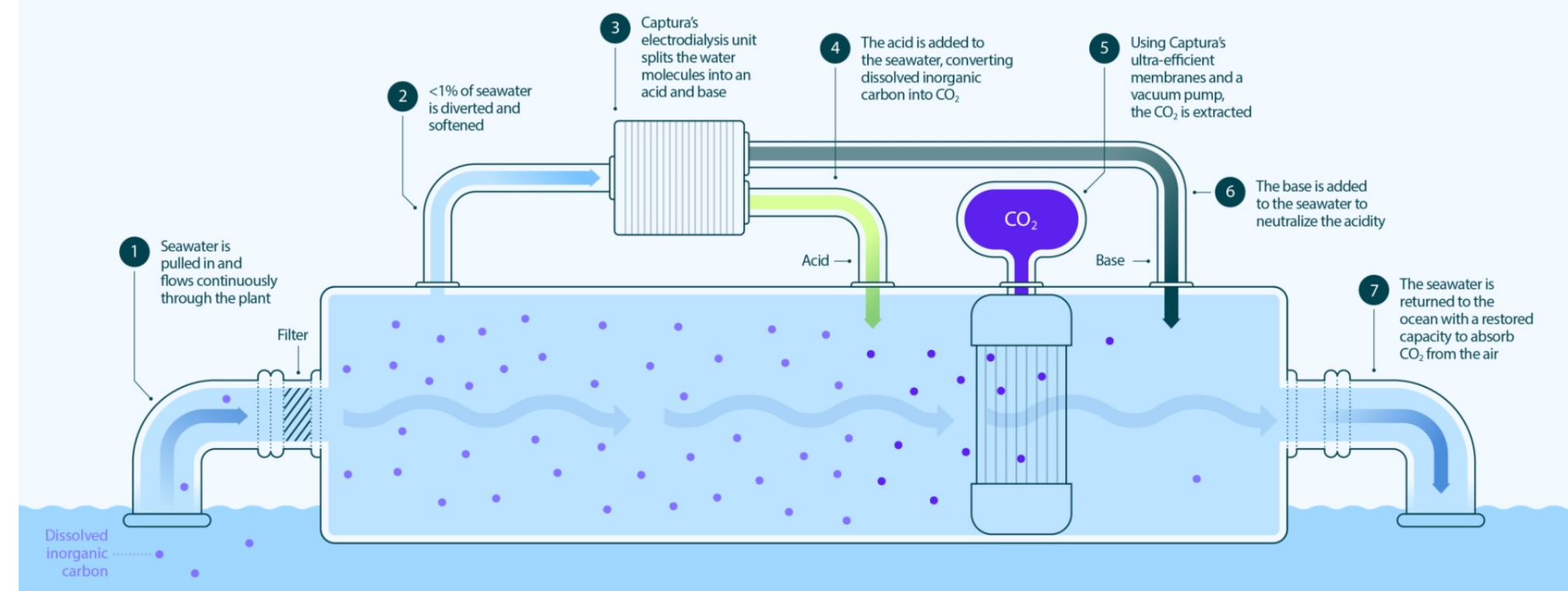


Fig 1. Captura's Direct Ocean Capture (DOC) process to extract CO₂ from seawater (top) and Captura's 100-tonne system at AltaSea at the Port of Los Angeles, CA (bottom).

Near Field Observational Data

A generalized near-field turbulent jet model is being developed and tested against field data from Captura's 100-ton pilot DOC system. The model incorporates water column profiling, Acoustic Doppler Current Profiler (ADCP), temperature chain data, and seawater carbonate system parameters from discrete samples.

Fig 2 (right). Surface pCO₂ data collection in the Port of Los Angeles within a range of ~10-100 meters. A 12-foot vessel equipped with a YSI EXO2 and a SparkFun RTK Express GNSS (with 1 cm accuracy) was utilized to create a pCO₂ distribution map.

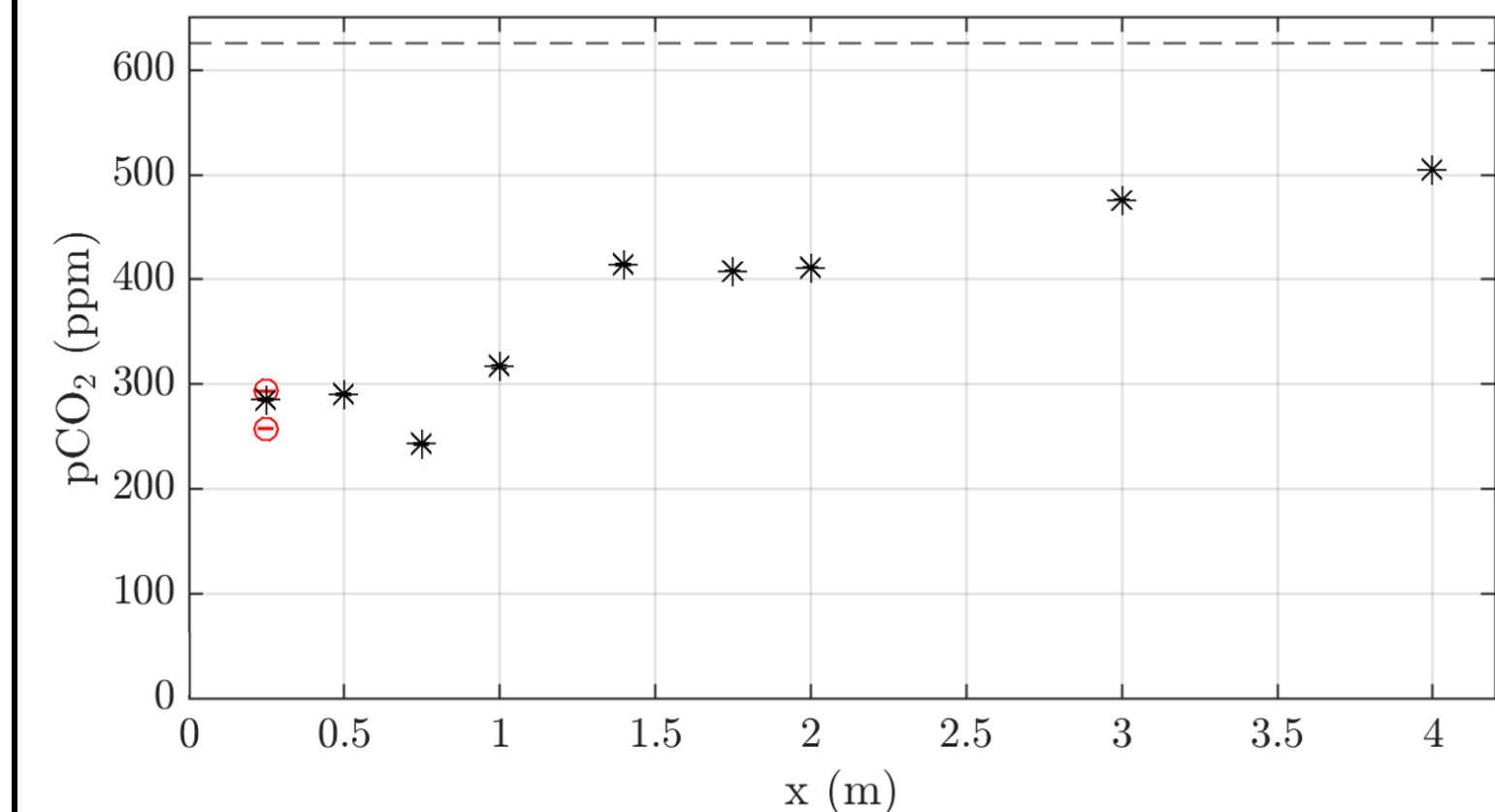
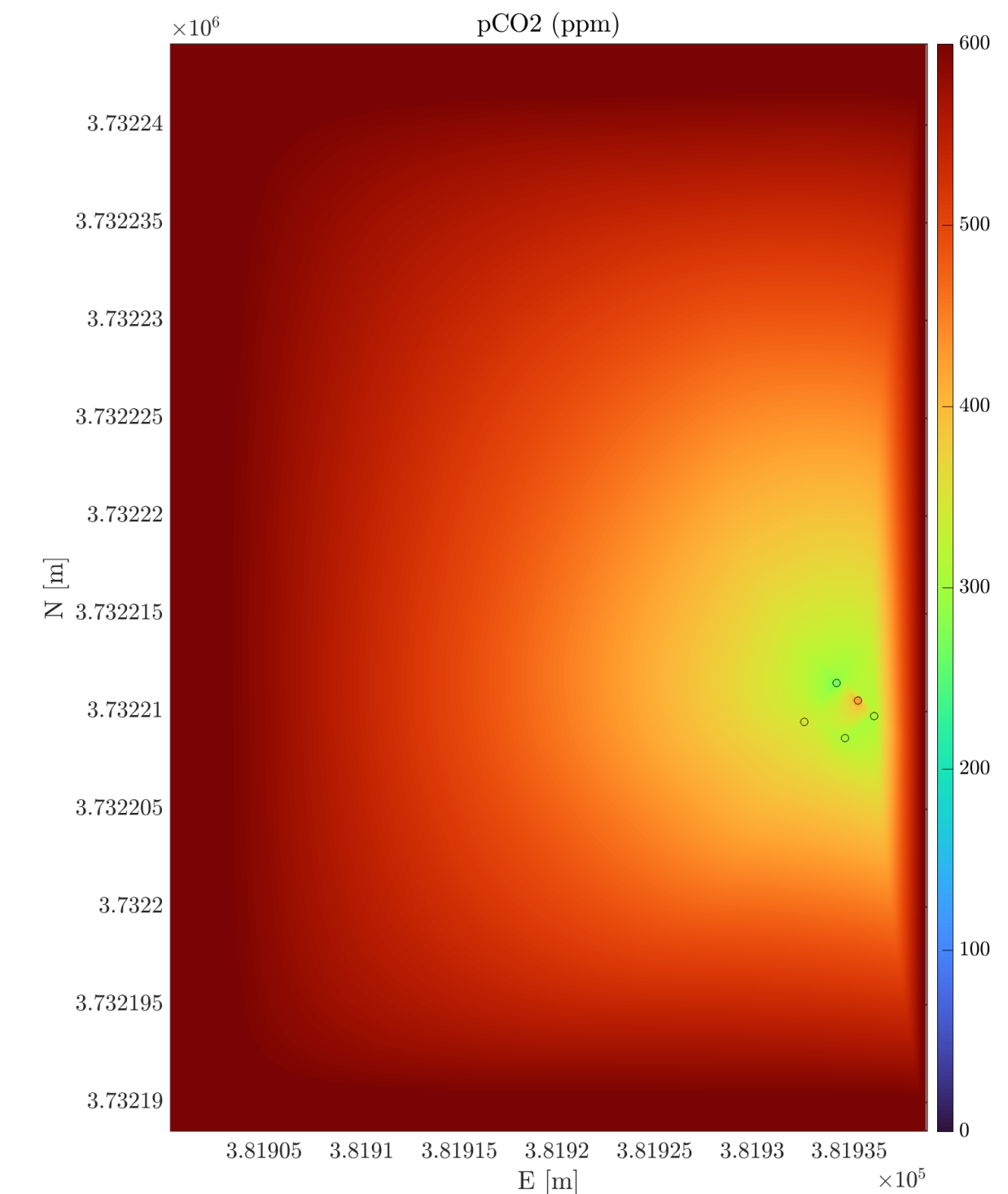


Fig 3 (left). Surface water samples were collected at multiple discharge points during Captura operations and pCO₂ levels were measured using a Pro-Oceanus NDIR sensor. The dashed line represents the baseline pCO₂, recorded away from the discharge with a separate in-situ pCO₂ sensor.

ROMS-BEC Parameterization

ROMS-BEC for DOC Components

- A horizontal grid cell resolution of 300m resolves coastal processes
- Hindcast simulation validated with 20 years of biogeochemical data in the Southern CA Bight
- Simulating DOC for a 20,000 tonnes CO₂ yr⁻¹ system and flow rate of 10 m³ s⁻¹
- Performing sensitivity tests for the spatial decoupling of the intake and discharge

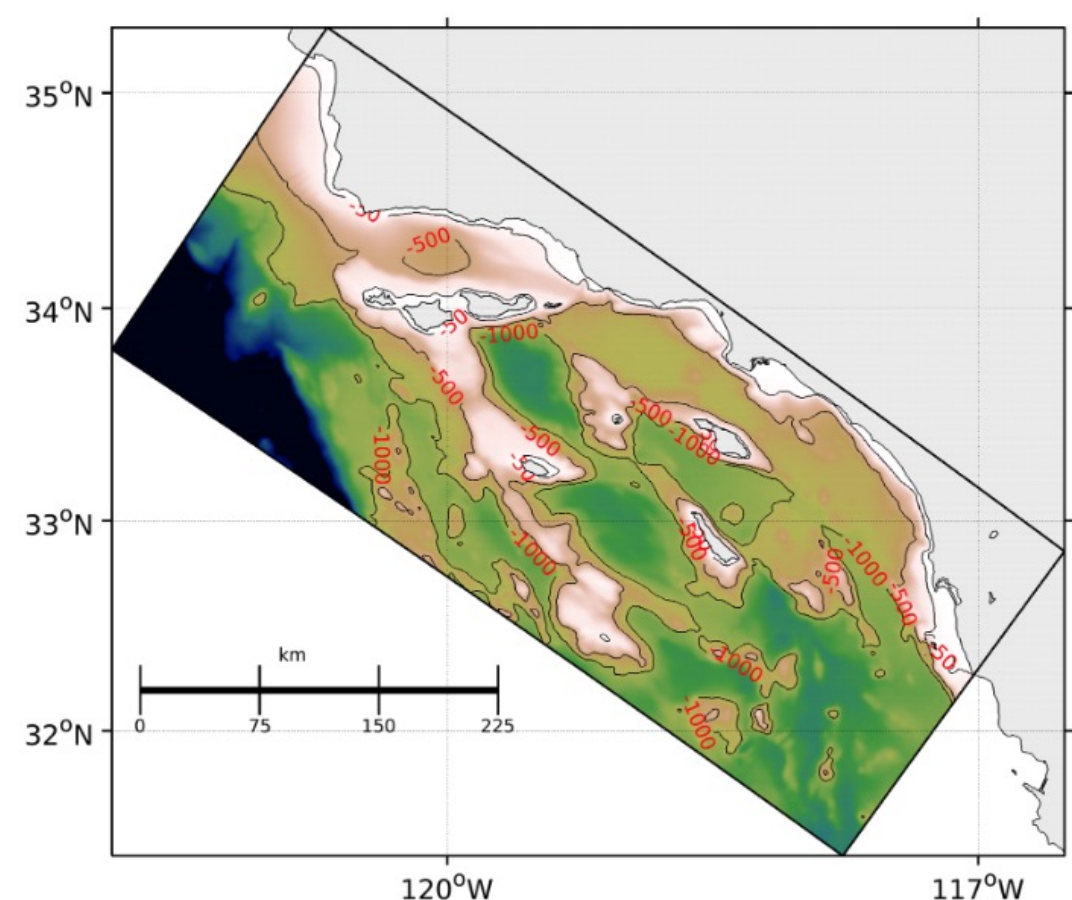
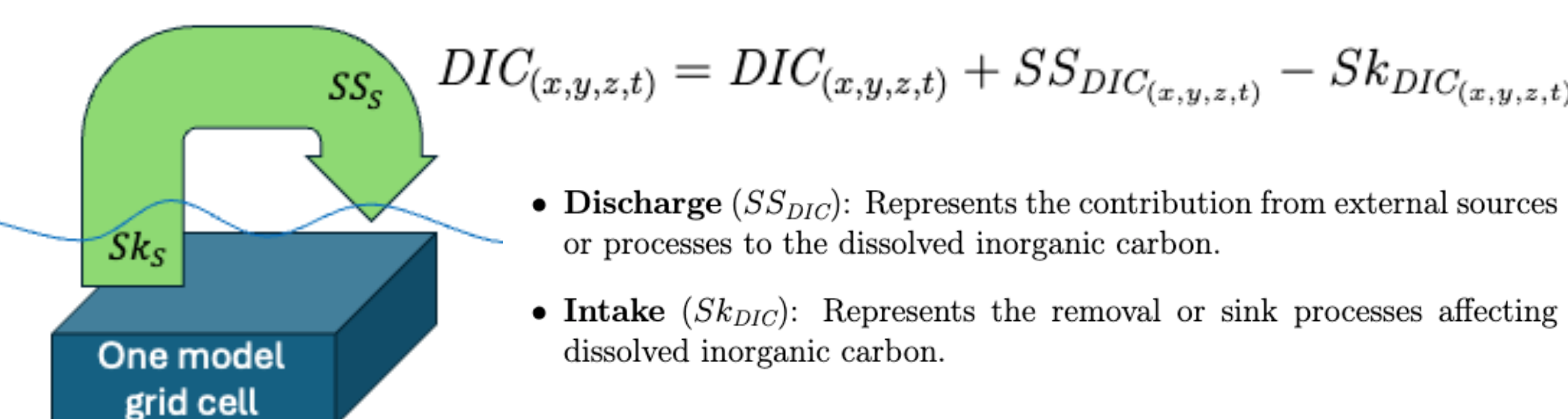


Fig 4. Model grid domain used to simulate the 20,000-tonne Captura plant. Seafloor depth (m) is represented by colormap and topographical contouring.



CO₂ Additionality Attributable to DOC

$$\Delta\Phi_{CO_2} = \Phi_{CO_2\text{-Direct Ocean Capture}} - \Phi_{CO_2\text{-Baseline}}$$

where Φ_{CO_2} is the air-sea gas exchange of CO₂ [mmol C m⁻² s⁻¹]

ROMS-BEC Determination of CO₂ Additionality

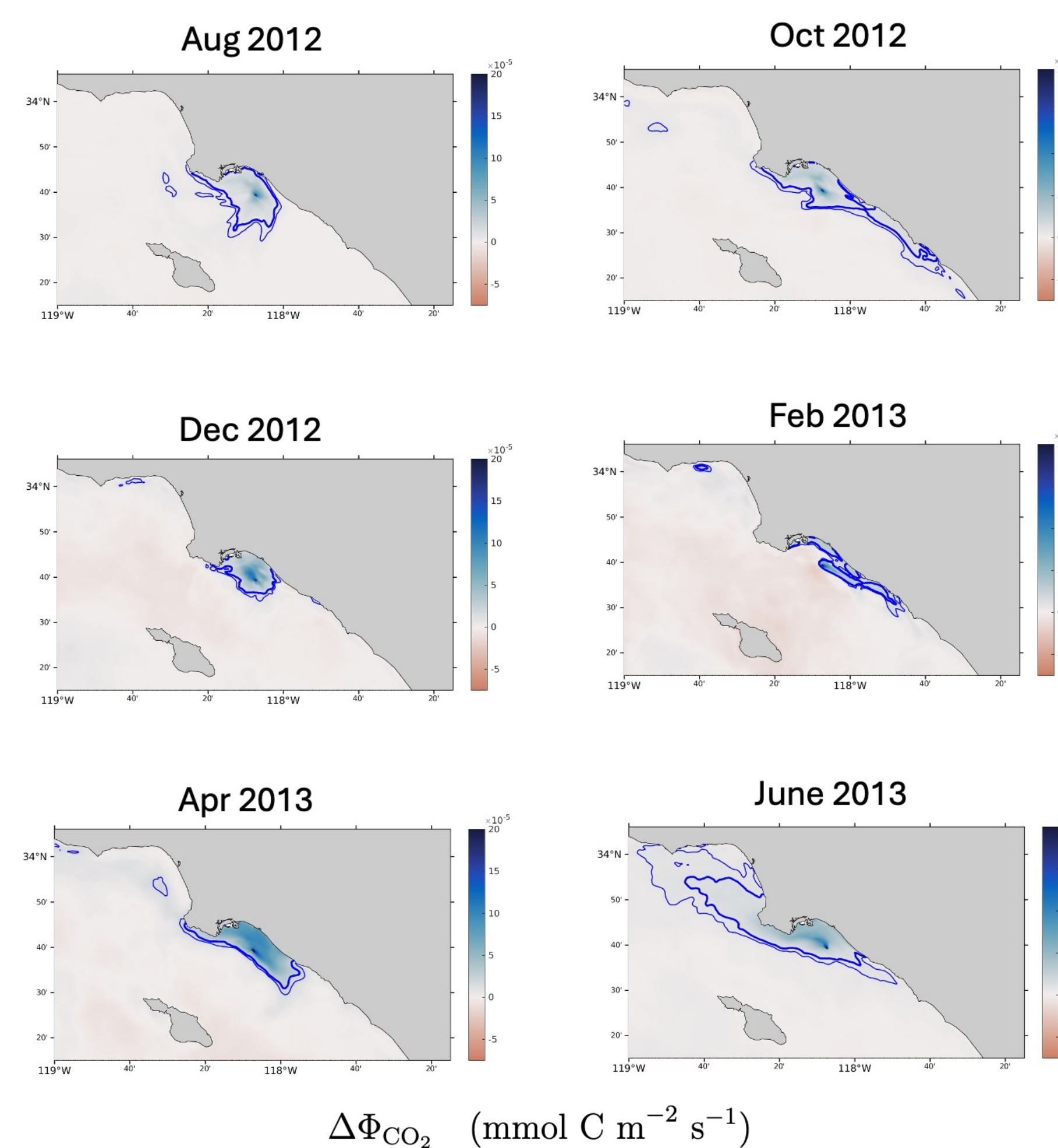


Fig 5. Time series showing difference in air-sea CO₂ exchange attributable to a DOC 20,000 tonne system relative to air-sea CO₂ exchange from a baseline scenario. Blue indicates an increase in the flux into the ocean. Contours define regions where changes in air-sea CO₂ exchange are attributable to DOC, exceeding intrinsic variability associated with the model. The extent and magnitude of the flux are influenced by seasonal variations in oceanographic conditions.

Preliminary Findings & Future Directions

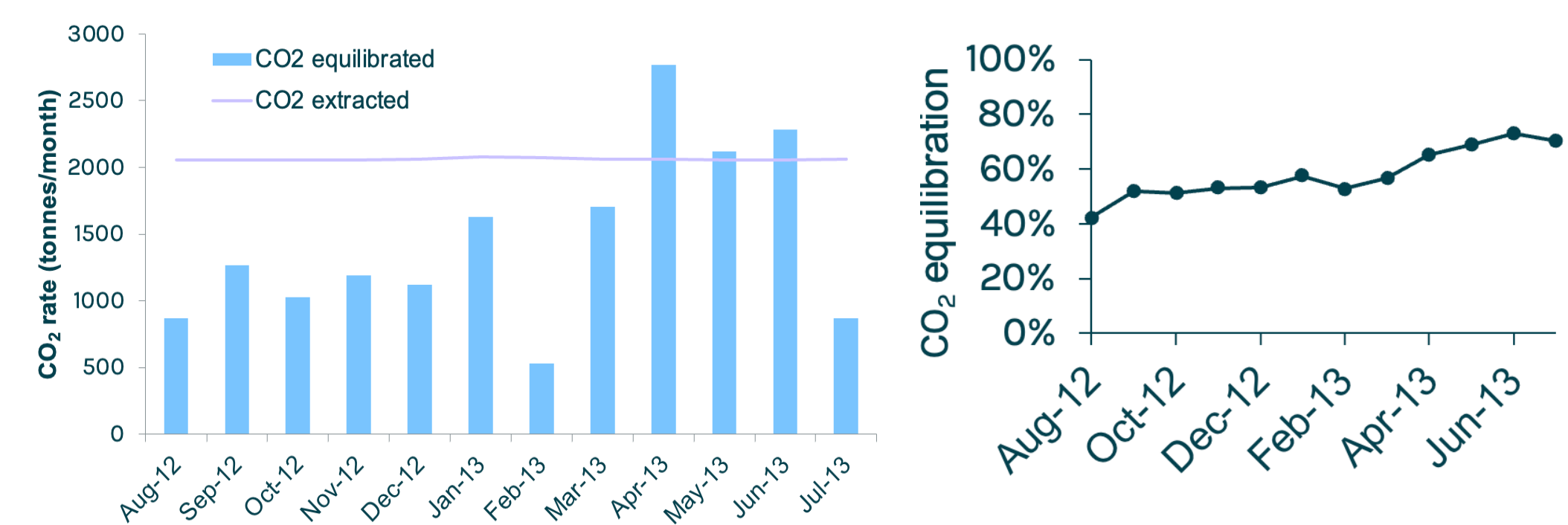


Fig 6. Projected CO₂ extraction (purple) and equilibration rates (blue) for Captura's model 20,000 tonne DOC system (left) and CO₂ air-sea equilibration of 70% within 1 year i.e. the drawdown of CO₂ from the atmosphere into the ocean reflects 70% of the CO₂ that was extracted during the DOC process (right).

Observational from plume discharge influencing plume dynamics-- primarily pH, DIC, temperature, and current velocity, can be used to refine and validate high resolution modeling. Such a model will simulate turbulent mixing, jet dispersion, and plume evolution across scales from meters to hundreds of meters. Calibrating the model with field data would enhance predictive accuracy for plume behavior under different discharge rates and environmental conditions. The next step would then be to couple near field model results with regional scale modeling. We aim to quantify the impact of DOC on the broader marine environment, refine carbon drawdown estimates, and establish frameworks for operational scalability and environmental compliance.

References and Acknowledgements

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